HCI for Sustainable and Mobile Application

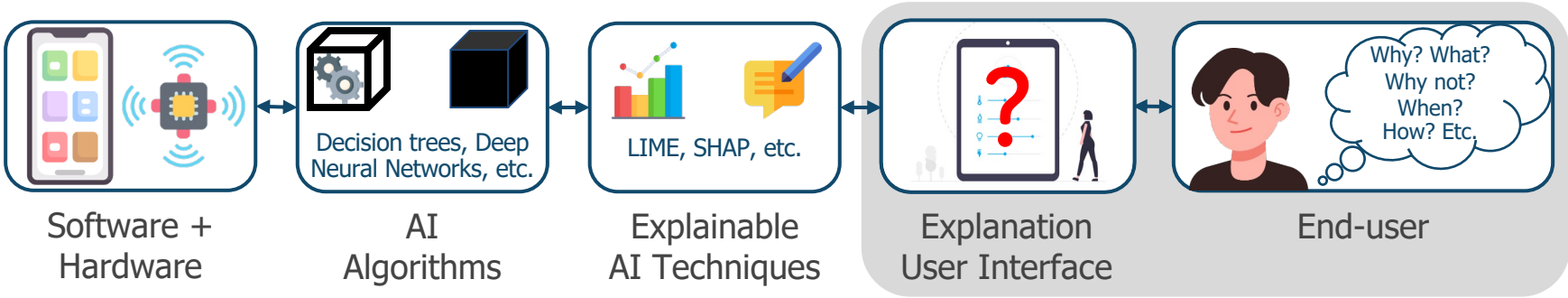
Explainable User Interfaces

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XAI System Overview

Explainable AI (XAI) system components [1]



Historical View [2]

Occasional interest in explanations of intelligent systems

- 1970s: Expert systems
- 1980s: Bayesian networks and artificial neural networks
- 2000s: Recommender systems

Recent interest in XAI

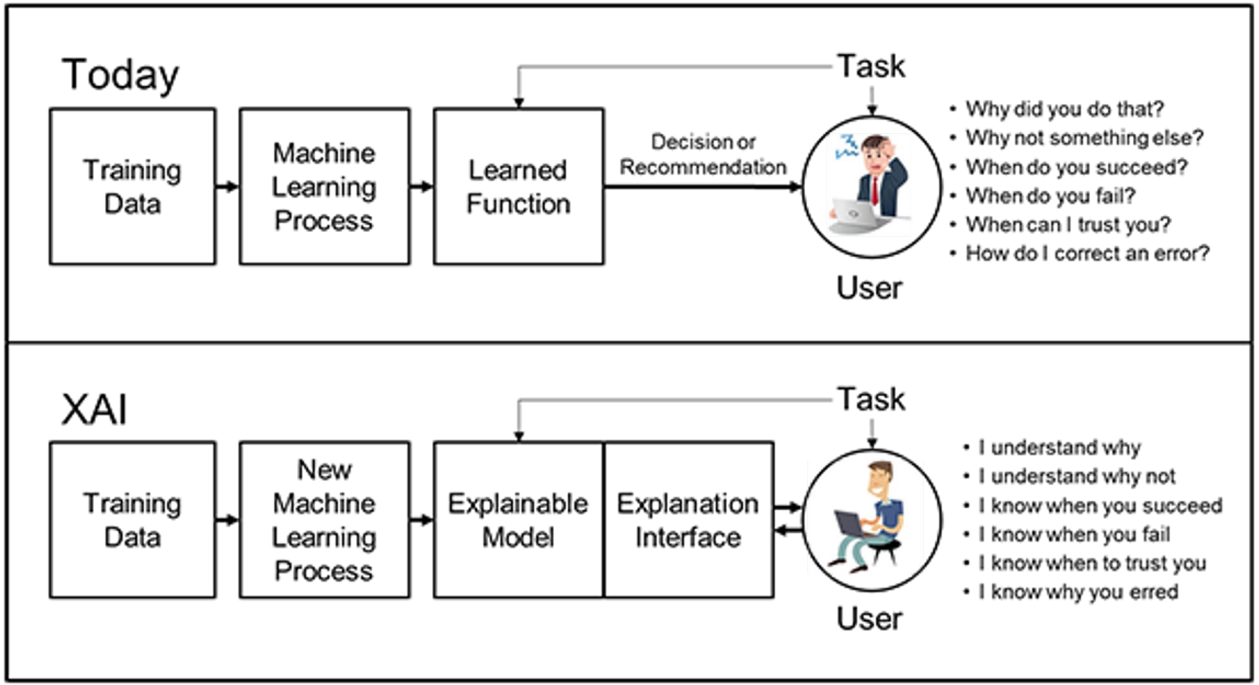
- 2018: Driven by DARPA[1] initiative to fund XAI

1 Defense Advanced Research Projects Agency

Historical View

- The success of Machine Learning and the use of complex and black-box algorithms (e.g. deep learning) calls for another wave of interest in Explainable Systems [2]
- Drivers of the explainable AI resurgence [3]:
  - Ethical concerns
  - Lack of *trust* on behalf of the end-users

Explainable Artificial Intelligence



https://www.darpa.mil/program/explainable-artificial-intelligence

Explainable Artificial Intelligence [2]

Perspective

New machine learning systems will have:

- ❑ The ability to explain their rationale
- ❑ Characterize their strengths and weaknesses
- ❑ Convey an understanding of how they will behave in the future

Strategy

- ❑ Develop new or modified ML techniques to produce more explainable models
- ❑ Combine these models with human-computer interaction techniques to translate them into understandable and useful explanation *dialogues* for the end-user

Related Concepts

Explainable AI

- "An explanatory agent revealing underlying causes to its or another agent's decision making" [3]

Explainability

- "How well a human could understand the decisions in the given context" [3]

Explanation

- "Explicitly explaining decisions to people" [3]

Related Concepts

Interpretability [4]

- "Interpret means to explain or to present in understandable terms"[2]
- In the context of ML systems, interpretability is defined as "the ability to explain or to present in understandable terms to a human"

2 Merriam-Webster dictionary

Related Concepts

FAT (Fairness, Accountability, and Transparency)

Fairness [5]

- Software fairness is a property of learning-based systems aimed to ensure that the software does not exhibit biases
- A fair software should not result in discriminatory outputs or behaviors for inputs relating to certain groups or individuals
- For instance, discriminatory inputs, inputs that are similar for a task but only differ in sensitive attributes (*e.g.*, gender, race or age), should produce similar outputs or induce similar program behaviors

Related Concepts

FAT (Fairness, Accountability, and Transparency)

Accountability [6]

- Any investigation must therefore consider algorithms as objects of human creation and take into account intent
- Including that of any group or institutional processes that may have influenced their design
- Personal view: It is closely related with social responsibility and justice

Related Concepts

FAT (Fairness, Accountability, and Transparency)

Key issues for algorithmic fairness and accountability are [2]:

- ❑ Bias
- ❑ Discrimination in big data and algorithms
- ❑ Ethics
- ❑ Legal and policy implications
- ❑ Data privacy
- ❑ Transparency

Related Concepts

FAT (Fairness, Accountability, and Transparency)

Transparency [7]

- Transparency aims to increase understanding and entails offering the user insight in how a system works, for example by offering explanations for system behaviour

Related Concepts

Intelligibility [8]

- "Context-aware systems that seek to act upon what they infer about the context must be able to represent to their users what they know, how they know it, and what they are doing about it"

Explanation Applications

AI sub-domains that cover explanation [3]:

- Justifying autonomous agent behaviour
- Debugging of ML models
- Explaining medical decision-making
- Explaining predictions of classifiers

Approaches and Toolkits [9]

Domains:

- Image classification [10]
- Text mining and text classification [11, 12, 13]
- Context-aware systems [8, 14]

White-box approaches: explaining the internal workings of the model [9]

- Visualization of node probabilities for Bayesian networks [15]
- Projection techniques for Support Vector Machines (SVMs) [16]
- Nomograms for SVMs [17]
- Graph visualization of neural networks [18]

Approaches and Toolkits [2]

Deep Neural Networks (DNNs):

- Grad-CAM: Visual explanations for Convolutional Neural Network (CNN)-based models [19]

Interpretable models [20]:

- Sparse linear classifiers
- Discretization methods: Decision trees and decision lists
- Prototype- or case-based classifiers
  - Nearest neighbors
  - Supervised optimization-based method

Approaches and Toolkits [9]

Black-box approaches: Explaining the relationships between input and output regardless of how complicated the model itself is [9]

- Prospector: Visual analytics system to understand how features affect prediction overall [21]
- Methods for visualizing how input features influence the outcome of neural network classifications [22, 23]
- Removal-based explanations for documents classification [12]:
  - "the classification would change to [name of an alternative class] if the words [list of words] were removed from the document"

Approaches and Toolkits [2]



An open-source, explainable AI toolkit built for analytics and autonomy applications

- Tools and resources to help users, developers, and researchers understand complex machine learning models
- Repository of independent contributions[3]

3 https://xaitk.org

Post-hoc XAI Techniques [24]

- Categories of techniques (black-box and interpretable models):
  - Global model explanations: Model's overall logic
  - Local explanations: Explaining a particular model output
  - Counterfactual explanations: Supports understanding how the model would behave with an alternative input
- Generated by techniques:
  - Feature-based explanations: Elucidate the model internals
  - Example-based explanations: Support case-based reasoning

Post-hoc XAI Techniques [24]

Global model explanations

- It is impossible to understand the internal workings of a black-box (opaque) model
- The aim is to provide an approximate overview of the model's behavior
- Training an interpretable model (e.g. decision tree, rule set or linear regression):
  - Same training data
  - Optimization to approximate the simple model's behavior to the original model
- Technique – Distillation: changes the learning objective of the interpretable model

Post-hoc XAI Techniques [24]

Local explanations

- Estimate the importance of each instance feature contributing to the prediction
- LIME (Local Interpretable Model-agnostic Explanations) algorithm:
  - Adds a small noise to the instance to create a set of neighbor instances
  - Fits a simple linear model on those neighbors that mimics the original model's behavior in the local region
  - Uses the linear model's weights as the feature importance to explain the prediction

Post-hoc XAI Techniques [24]

Local explanations

- Estimate the importance of each instance feature contributing to the prediction
- SHAP (SHapley Additive exPlanations) algorithm:
  - Feature importance: Based on Shapley values to assign credit to each feature
  - Inspired by cooperative game theory
  - Explain by visualizing the feature-importance
  - Explain by describing the most important feature for the prediction

Post-hoc XAI Techniques [24]

Local explanations

- Estimate the importance of each instance feature contributing to the prediction
- For Deep Neural Networks:
  - There are several algorithms to identify important parts of input features based on gradient, propagation, occlusion, etc.
  - Also referred as saliency methods
  - Generate saliency maps when applied to image data

Post-hoc XAI Techniques [24]

Local explanations

- Estimate the importance of each instance feature contributing to the prediction
- Example-based method:
  - Notion of similarity
  - Justify the prediction by finding similar instances in the training data with the same predicted outcome

Post-hoc XAI Techniques [24]

Counterfactual explanations: ”Counter to the facts”

- How the model would behave when the current input changes
- "Why not a different prediction" or "how to change to get a different prediction" questions rather than a descriptive "why" question
- Identify minimum changes for a instance to receive a different prediction
- Counterfactual explanations are also referred as contrastive explanations for a counterfactual outcome

Post-hoc XAI Techniques [24]

Counterfactual explanations: ”Counter to the facts”

- How the model would behave when the current input changes
- Example-based method:
  - Generate counterfactual examples: Instances with minimal differences compared to the original one but having a different outcome
- "What if" question: Explore a specific feature on how its changes impact the model's prediction
  - Feature inspection techniques: Partial Dependence Plot (PDP) and Individual Conditional Expectation (ICE)

User Groups and Explanations [24]

- Model developers: To improve or debug the model
- Business owners or administrators: To evaluate an AI application's capability, regulatory compliance, etc. to determine its adoption and usage
- Decision-makers (users of AI decision support applications): To form appropriate trust in the AI and make informed decisions
- Impacted groups: To seek recourse or contest the AI
- Regulatory bodies: To audit for legal or ethical compliance such as fairness, safety, privacy, etc.

One person can assume different roles and needs at certain usage times

XAI Question Categories [24]

- **How:** The general logic or process the AI follows to have a global view
- **Why:** The reason behind a specific prediction
- **Why Not:** Why the prediction is different from an expected or desired outcome
- **How to be That:** Ways to change the instance to get a different prediction
- **How to Still Be This:** What change is allowed for the instance to still get the same prediction
- **What if:** How the prediction changes if the input changes
- **Performance:** The AI's performance
- **Data:** The training data
- **Output:** What can be expected or done with the AI's output

Questions and XAI Methods [24]

Question	Ways to explain	Example XAI methods
How (global model-wide)	• Describe the general model logic as feature impact*, rules† or decision-trees‡ • If user is only interested in a high-level view, describe what are the top features or rules considered	ProfWeight*†‡ [28], Global feature importance* [71, 105], Global feature inspection plots* (e.g. PDP [49]), Tree surrogates‡ [25]
Why (a given prediction)	• Describe how features of the instance, or what key features, determine the model's prediction of it* • Or describe rules that the instance fits to guarantee the prediction† • Or show similar examples with the same predicted outcome to justify the model's prediction‡	LIME* [89], SHAP* [72], LOCO* [63], Anchors† [90], ProtoDash‡ [47]
Why Not (a different prediction)	• Describe what features of the instance determine the current prediction and/or with what changes the instance would get the alternative prediction* • Or show prototypical examples that have the alternative outcome†	CEM* [27], Counterfactuals* [69], ProtoDash† (on alternative prediction) [47]
How to Be That (a different prediction)	• Highlight feature(s) that if changed (increased, decreased, absent, or present) could alter the prediction to the alternative outcome, with minimum effort required* • Or show examples with minimum differences but had the alternative outcome†	CEM* [27], Counterfactuals* [69], Counterfactual instances† [100], DiCE† [78]
How to Still Be This (the current prediction)	• Describe features/feature ranges* or rules† that could guarantee the same prediction • Or show examples that are different from the instance but still had the same outcome	CEM* [27], Anchors† [90]
What if	• Show how the prediction changes corresponding to the inquired change of input	PDP [49], ALE [10], ICE [44]
Performance	• Provide performance information of the model • Provide uncertainty information for each prediction* • Describe potential strengths and limitations of the model	Precision, Recall, Accuracy, F1, AUC; Communicate uncertainty of each prediction* [42]; See examples in FactSheets [11] and Model Cards [77]
Data	• Provide comprehensive information about the training data, such as the source, provenance, type, size, coverage of population, potential biases, etc.	See examples in FactSheets [11] and Datasheets [39]
Output	• Describe the scope of output or system functions. • If applicable, suggest how the output should be used for downstream tasks or user workflow	See examples in FactSheets [11] and Model Cards [77]

Explanation Formats

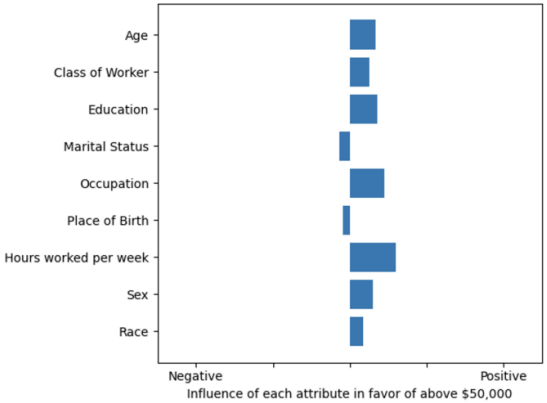
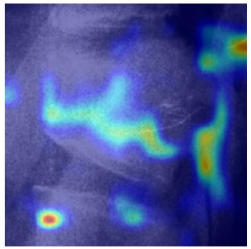


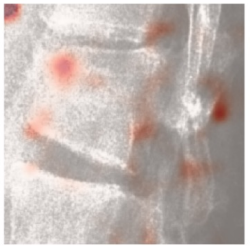
Chart – Feature contribution [26]



_ has a degree in education from Kent State University with a major in hearing impaired, Special Education, K-12. _ has also done work in the Masters of Education Program at Kent State. _ has been involved in the preschool as a parent and has many years of teaching experience working with young children.

Textual + Color scale [26]





Saliency-based highlights [27]

XAI Software Development [25]



Requirements Analysis	Design	Implementation	Evaluation	Deployment
What requirements need to be gathered and analyzed to select suitable XAI methods?	What types of explanations exist and how can they be effectively delivered to the end user?	What are current XAI implementations and libraries?	How to perform reliable evaluations of XAI software?	What can we learn about the deployment of XAI software?

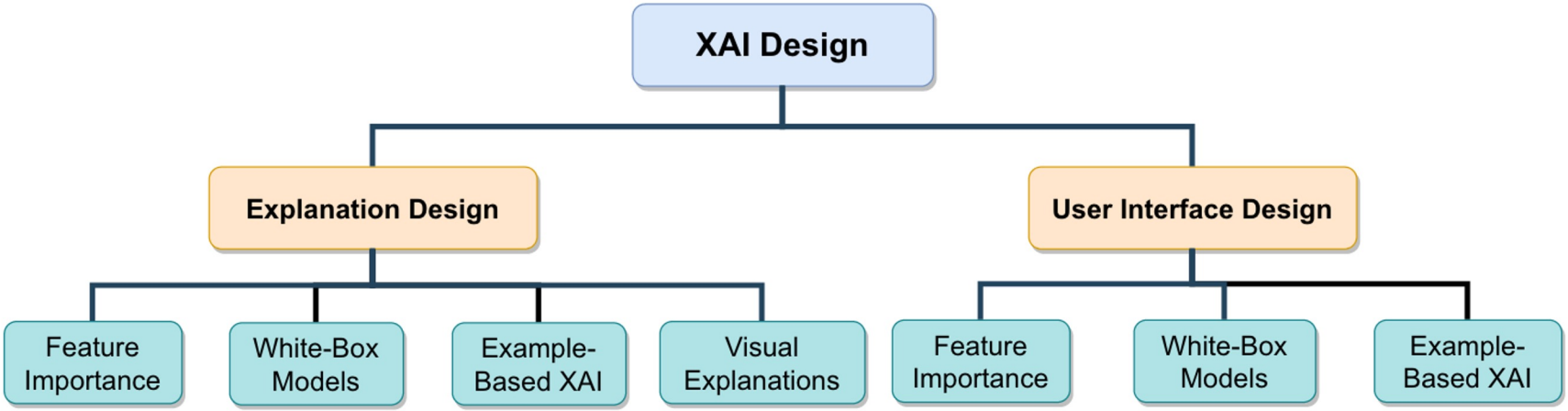
Requirement Analysis [25]

- Specify what needs to be explained and to whom
  - Explore user characteristics: AI knowledge, domain knowledge, attitude toward AI, responsibilities, cognitive abilities
- Specify how to explain
- Specify the most frequently used data type of an application: Images, tabular data
- Elicit requirements from the underlying ML model (architecture)
- Elicit requirements based on the existence of an AI system for which the explanations are needed (post-hoc techniques)

Requirement Analysis [25]

- Elicit requirements based on decision characteristics: Outcome criticality, time sensitivity, decision complexity
- Elicit requirements from the context of explanations (constraints)
- Trade-off analysis

XAI Design [25]



Explanation Design [25]

Explanation Type	Black-Box Model	Method	Scope	Functionality
Feature Importance	Any	LIME	Local	Surrogate Model
		LORE	Local	Surrogate Model
		Anchors	Local	Surrogate Model
		Occlusion	Local	Input Perturbation
		Permutation Feature Importance	Global	Input Perturbation
		Shapley Feature Importance	Global	Game-Theory
		SHAP	Both	Game-Theory
	Neural Network	Guided Backpropagation	Local	Backpropagation
		Integrated Gradients	Local	Backpropagation
		Layerwise Relevance Propagation	Local	Backpropagation
		DeepLift	Local	Backpropagation
		Testing with Concept Activation Vectors	Global	Human Concepts
		Activation Maximization	Global	Forwardpropagation
	CNN	Deconvolution	Local	Backpropagation
		Class Activation Map	Local	Backpropagation
		Grad-CAM	Local	Backpropagation
	Transformer	Attention Flow/Attention Rollout	Local	Network Graph
		Transformer Relevance Propagation	Local	Backpropagation
White-Box Model	Any	Rule Extraction	Global	Simplification
		Tree Extraction	Global	Simplification
		Model Distillation	Global	Simplification
	CNN	Attention Network	Global	Model Adaption
	RNN	Attention Network	Global	Model Adaption
Example-Based	Any	Prototypes	Global	Example (Train Data)
		Critisisms	Global	Example (Train Data)
		Counterfactuals	Global	Fictional data point
Visual Explanations	Any	Partial Dependence Plot	Global	Marginalization
		Individual Conditional Expectation	Global	Marginalization
		Accumulated Local Effects	Global	Accumulation

Explainable User Interface (XUI) Design [25]

- Characteristics of human-friendly explanations:
  - Contrastiveness
  - Truthfulness
  - Selectiveness
- Design principles for XUI:
  1. Combine texts and images to facilitate understanding and communicate effectively
  2. Offer hierarchical or iterative functionality that efficiently allows follow-ups on initial explanations
  3. Multiple explanation methods and modalities can help to triangulate insights
  4. Adjust explanations to the user's mental model and context

Explainable User Interface (XUI) Design [25]

Feature importance

- Heatmaps for image data
  - SmoothGrad to reduce noise in heatmaps
- Highlight individual tokens for text data
- Interactive visualization of the self-attention for transformer-based architectures
- Tornado diagrams for structured data

Explainable User Interface (XUI) Design [25]

Feature importance

- Textual explanations from feature importance vectors for recommender systems
- Template-based approaches with presets of sentences based on the feature importance values
- Generate textual explanations through a neural network
- Generator network whose explanations are restricted by a surrounding template

Explainable User Interface (XUI) Design [25]

White-box models

- Feature importance approaches can be applied to white-box models
- Decision tables or rule sets in textual form for decision rules
  - Sankey diagrams and visualizations of data distributions

Explainable User Interface (XUI) Design [25]

Example-based XAI

- Inherently comprehensible for humans
- Combine prototypes with a local approach

Additional Courses

- **Codecademy – Learn Explainable AI**

  https://www.codecademy.com/learn/learn-explainable-ai

- **Coursera – Explainable AI (XAI) Specialization**

  https://www.coursera.org/specializations/explainable-artificial-intelligence-xai

Implementation Phase [25]

Python libraries

- Captum: Feature importance methods for neural networks implemented in Pytorch
- DiCE: Generation of counterfactuals
- PAIR Saliency: Various methods to generate feature importance maps for images
- Quantus: Quantitative evaluation of explanations for image classification (metrics)

Implementation Phase

Python libraries

LIME

```
pip install lime
```

```
import lime
import lime.lime_tabular

explainer = lime.lime_tabular.LimeTabularExplainer(training_data,
mode="classification", feature_names=feature_names)
explanation = explainer.explain_instance(test_data, model.predict_proba)
explanation.show_in_notebook()
```

https://spotintelligence.com/2024/01/15/explainable-ai/

Implementation Phase

Python libraries

SHAP

```
pip install shap
```

```
import shap

explainer = shap.Explainer(model)
shap_values = explainer.shap_values(test_data)
shap.summary_plot(shap_values, test_data)
```

https://spotintelligence.com/2024/01/15/explainable-ai/

Implementation Phase

Python libraries

InterpretML

```
pip install interpret
```

```
from interpret import show
from interpret.data import ClassHistogram
from interpret.glassbox import ExplainableBoostingClassifier

ebm = ExplainableBoostingClassifier()
ebm.fit(X_train, y_train)

show(ClassHistogram(ebm, feature_names=feature_names))
```

https://spotintelligence.com/2024/01/15/explainable-ai/

Implementation Phase

Python libraries

ELI5 (Explain Like I'm 5)

- For debugging and visualizing ML models

```
pip install eli5
```

```
import eli5

eli5.show_weights(model)
```

https://spotintelligence.com/2024/01/15/explainable-ai/

Implementation Phase

Python libraries

Yellowbrick

- Visualization library for ML diagnostics and interpretability

```
pip install yellowbrick
```

```
from yellowbrick.model_selection import FeatureImportances

viz = FeatureImportances(model)
viz.fit(X_train, y_train)
viz.show()
```

https://spotintelligence.com/2024/01/15/explainable-ai/

Implementation Phase [25]

Popular XAI libraries

Name	Focus	Feature Importance	White-Box Models	Example-Based XAI	Visual XAI	Framework
AIX 360 [151]	General	LIME, SHAP	Decision Rules, Model Distillation	Prototypes, Contrastive Explanations	—	—
Alibi [152]	General	Anchors, Integrated Gradients, SHAP,	—	Contrastive Explanations, Counterfactuals	ALE	TensorFlow
Captum [153]	Neural Networks	DeepLift, Deconvolution, Integrated Gradients, SHAP, Guided Backpropagation, GradCam, Occlusion, PFI	—	—	—	PyTorch
DALEX [154]	General	LIME, SHAP, PFI	—	—	ALE, PDP	—
DiCE [155]	Counterfactuals	—	—	Counterfactuals	—	—
InterpretML [156]	General	LIME, SHAP, Morris Sensitivity Analysis	Explainable Boosting, Decision Tree, Decision Rules, Regression	—	PDP	—
PAIR Saliency [157]	Saliency Maps	Integrated Gradients, GradCam, Occlusion, Guided Backpropagation, Ranked Area Integrals, SmoothGrad	—	—	—	PyTorch, TensorFlow
Skater [158]	General	Layerwise Relevance Propagation, LIME, Integrated Gradients, Occlusion, PFI	Bayesian Rule List, Decision Tree	—	PDP	TensorFlow
Quantus [159]	Quantitative Evaluation	—	—	—	—	TensorFlow, PyTorch
ExplainerDashboard [160]	General	SHAP, PFI	Decision Tree	—	PDP	Scikit-learn
Ecco [161]	NLP	Integrated Gradients, Saliency, DeepLift, Guided Backprop	—	—	—	PyTorch
XAITK [162]	General	Saliency Maps	Decision Tree	Explanation by Example	—	—

Evaluation Phase [25]

Computational Evaluation (Metrics)

- Robustness: Similarity scores, sensitivity analysis, and the variation between an explanation obtained from the original input and its perturbed version
- Faithfulness: Most strategies rely on adding noise to the focal variable as a mean to remove the information it contains

Evaluation Phase [25]

Computational Evaluation (Metrics)

- Complexity:
  - Number of important features on which the model relies for its prediction
  - Importance distribution onto the entire set of features
  - Gini-index of the feature attribution vector as a measurement of its sparseness
  - Entropy for fractional attribution of each feature related to the total attribution
  - Number of features that exceed a specific attribution threshold
- Task-specific evaluation: Class sensitivity, localization metric for image classification

Evaluation Phase [25]

Computational Evaluation (Metrics)

- Method-specific evaluation:
  - Fidelity: Describes how well a white-box model matches the black-box model
  - Proximity: Describes the similarity of the test instance to the generated counterfactual
  - Sparsity: Describes the number of changed features in the counterfactual example
  - Relative distance: Compares the mean distance of the generated counterfactuals to the mean distance of counterfactual instances
- Sanity Checks: Model parameter randomization test and the data randomization test

Evaluation Phase [25]

Human-based evaluation:

- Goodness: Attributes used by consensus to describe a useful explanation
- User satisfaction: Contextualized measurement considering whether an explanation is adequate in the user's situation
  - "Degree to which users feel that they understand the AI model or process being explained to them"
- Mental model: Level of understanding of the underlying AI model

Deployment Phase [25]

Python package `explainerdashboard`:

- Quick deployment of a Web-based interpretability dashboard that explains the workings of traditional ML models

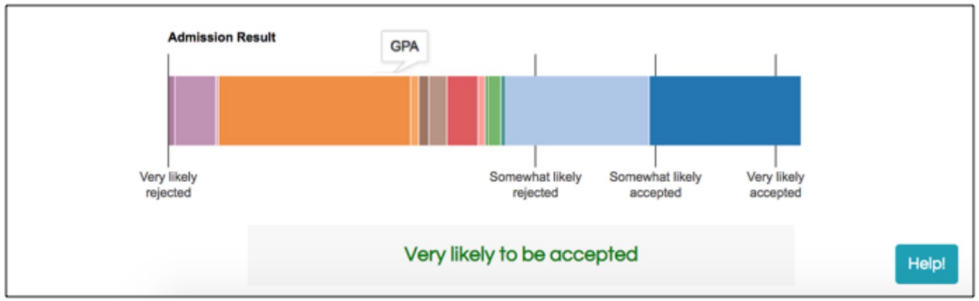
Explanation Interfaces [9]

Black-box vs. white-box

Black-box

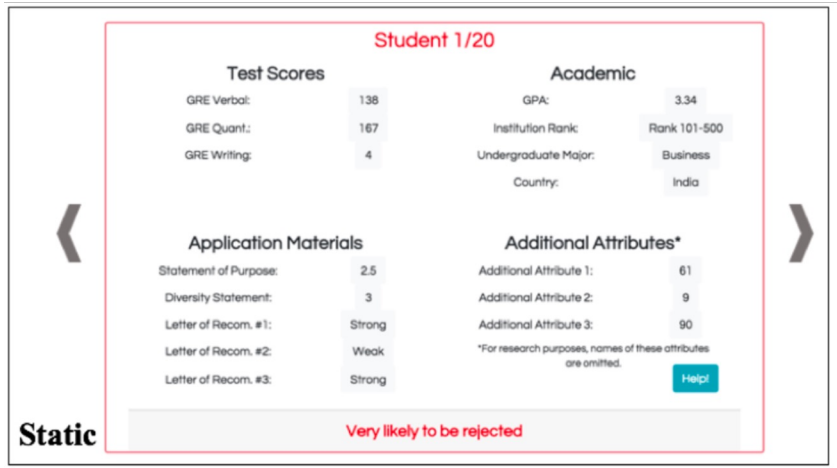


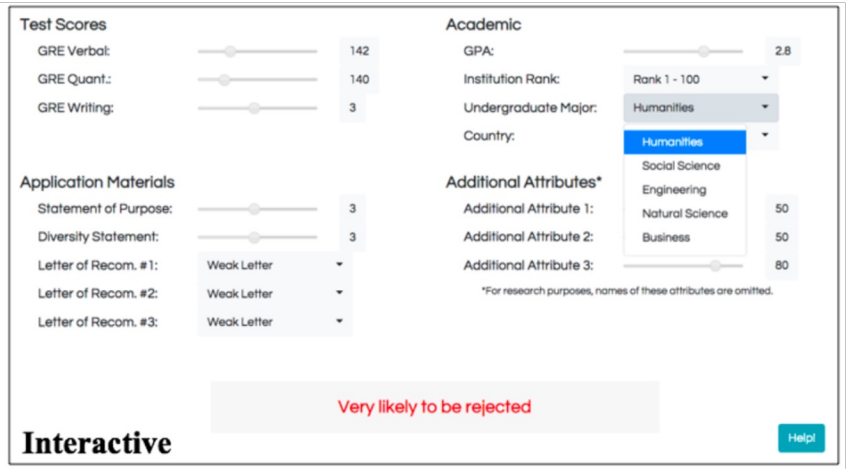
White-box



Explanation Interfaces [9]

Static vs. interactive





Explanation Interfaces [9]

Interactive vs. static interfaces

- Interactive interfaces increased users' understanding of the algorithm compared to the static ones
- Users using the interactive interfaces spent more time answering the questions

White-box vs. black-box interfaces

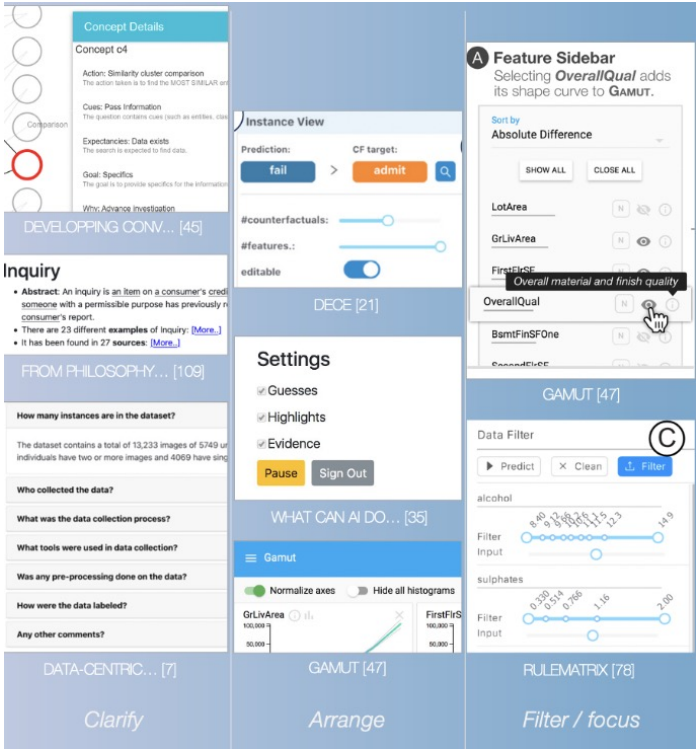
- White-box interfaces increased users' objective understanding compared to the black-box interfaces
- White-box interfaces reveal more complexity and introduce cognitive workload

Interactive Explanations [28]

Taxonomy of XAI interactivity techniques

Selective Explanations – Tasks:

- ❑ Clarify: Give additional information/explanations on demand
- ❑ Arrange: Choose and organize the explanation type(s), parameters and visual representation(s)
- ❑ Filter/Focus: Filter the explanation according to an input/input metric

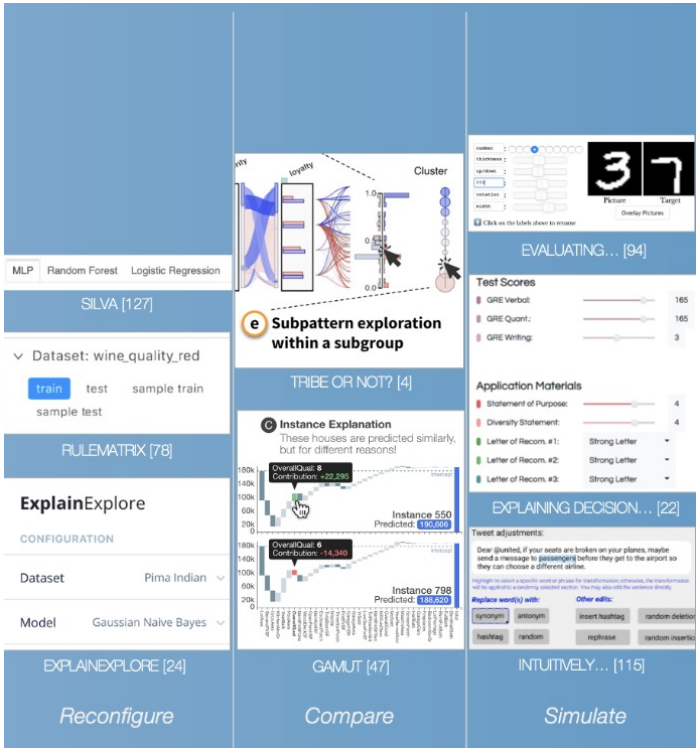


Interactive Explanations [28]

Taxonomy of XAI interactivity techniques

Mutable Explanations – Tasks:

- Reconfigure: Change the dataset or the AI model (parameters) and show me the corresponding prediction and explanations
- Simulate: Change the inputs, the output or the dataset distribution and show me the corresponding prediction and explanations
- Compare: Show me explanations of related or selected data inputs or outputs

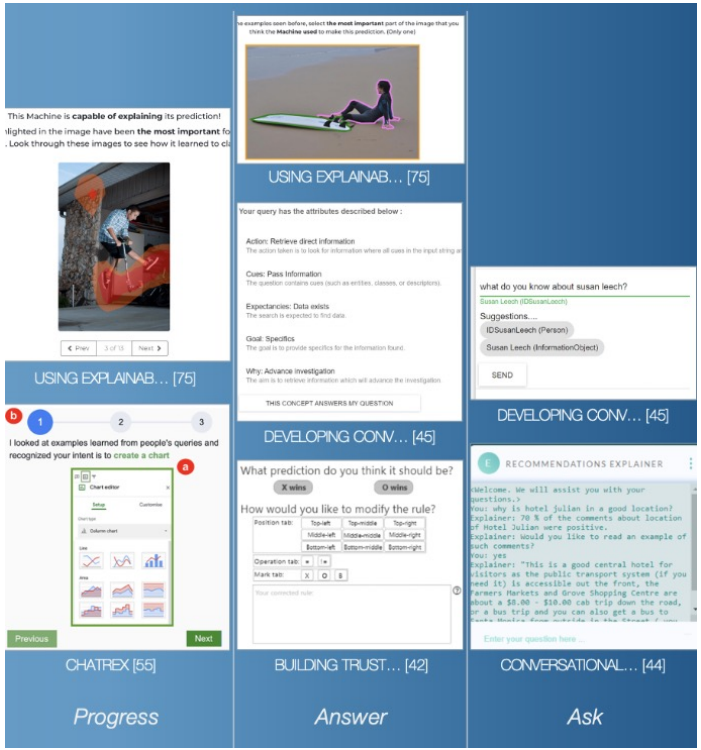


Interactive Explanations [28]

Taxonomy of XAI interactivity techniques

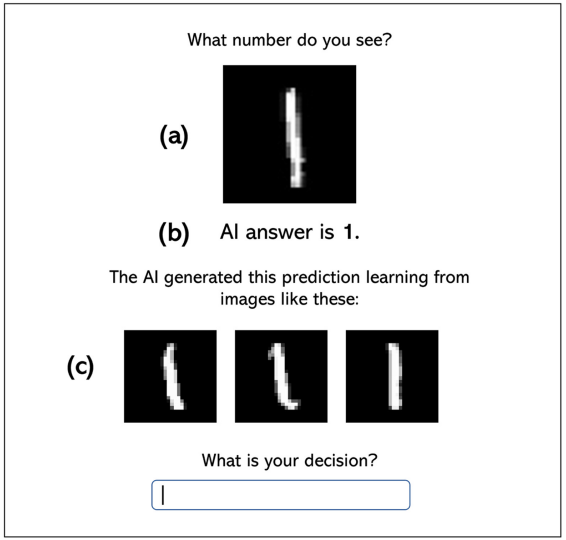
Dialogic Explanations – Tasks:

- Progress: Guide user through an explanation sequence
- Answer: Give feedback, edit explanation components
- Ask: Iterative questions and receive answers following a dialogue structure

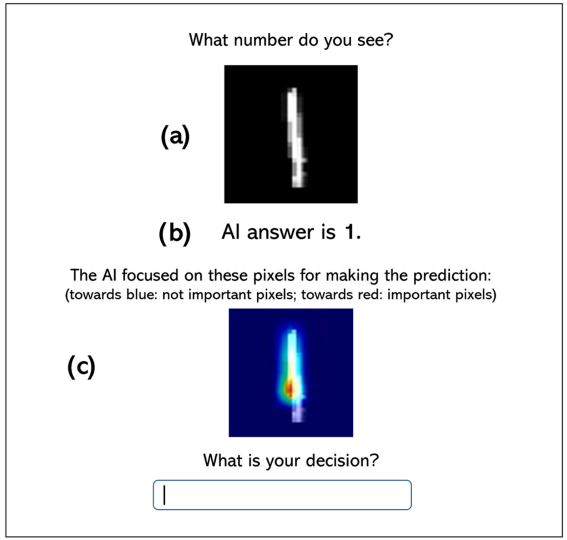


Example Explainable User Interfaces

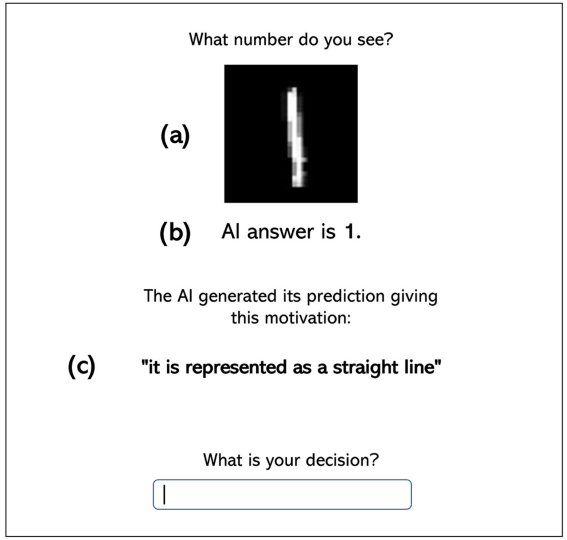
Letter Identification [29]



(1)

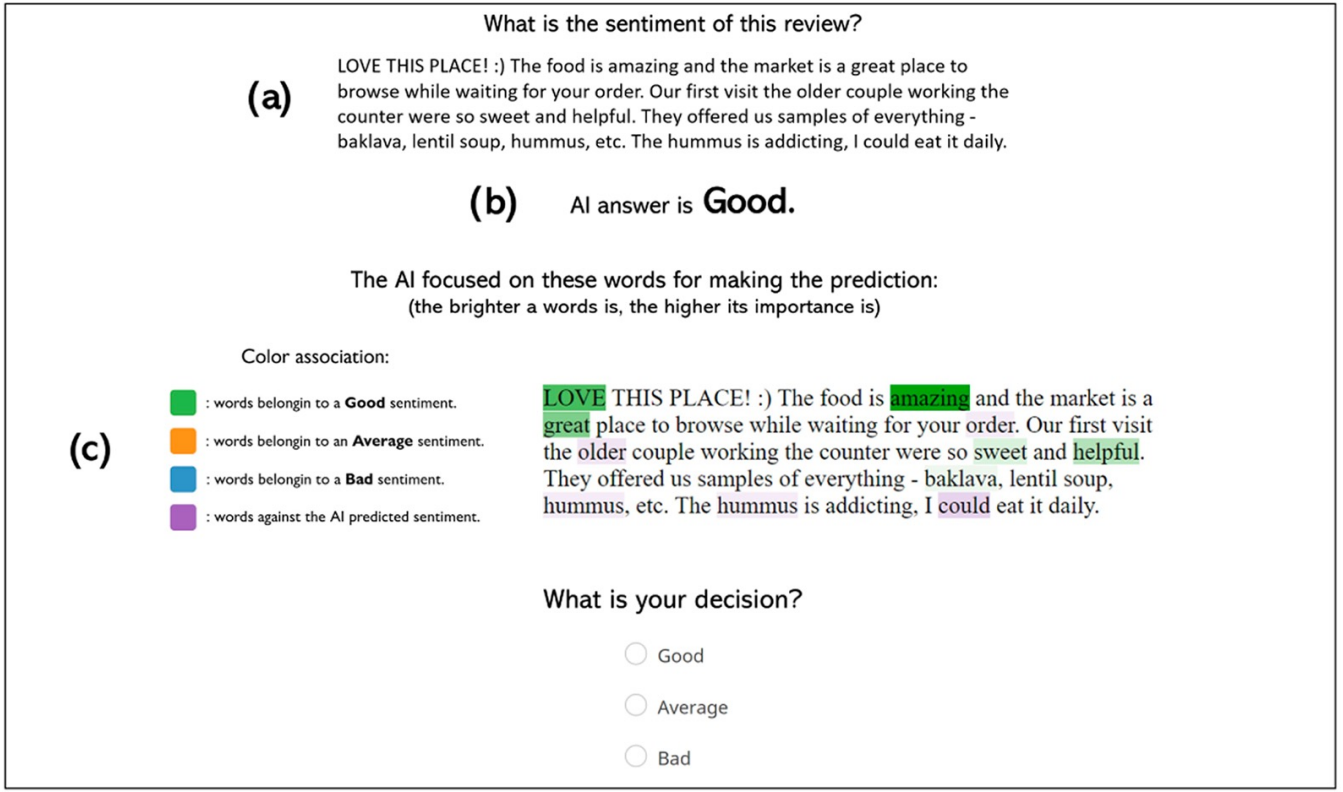


(2)

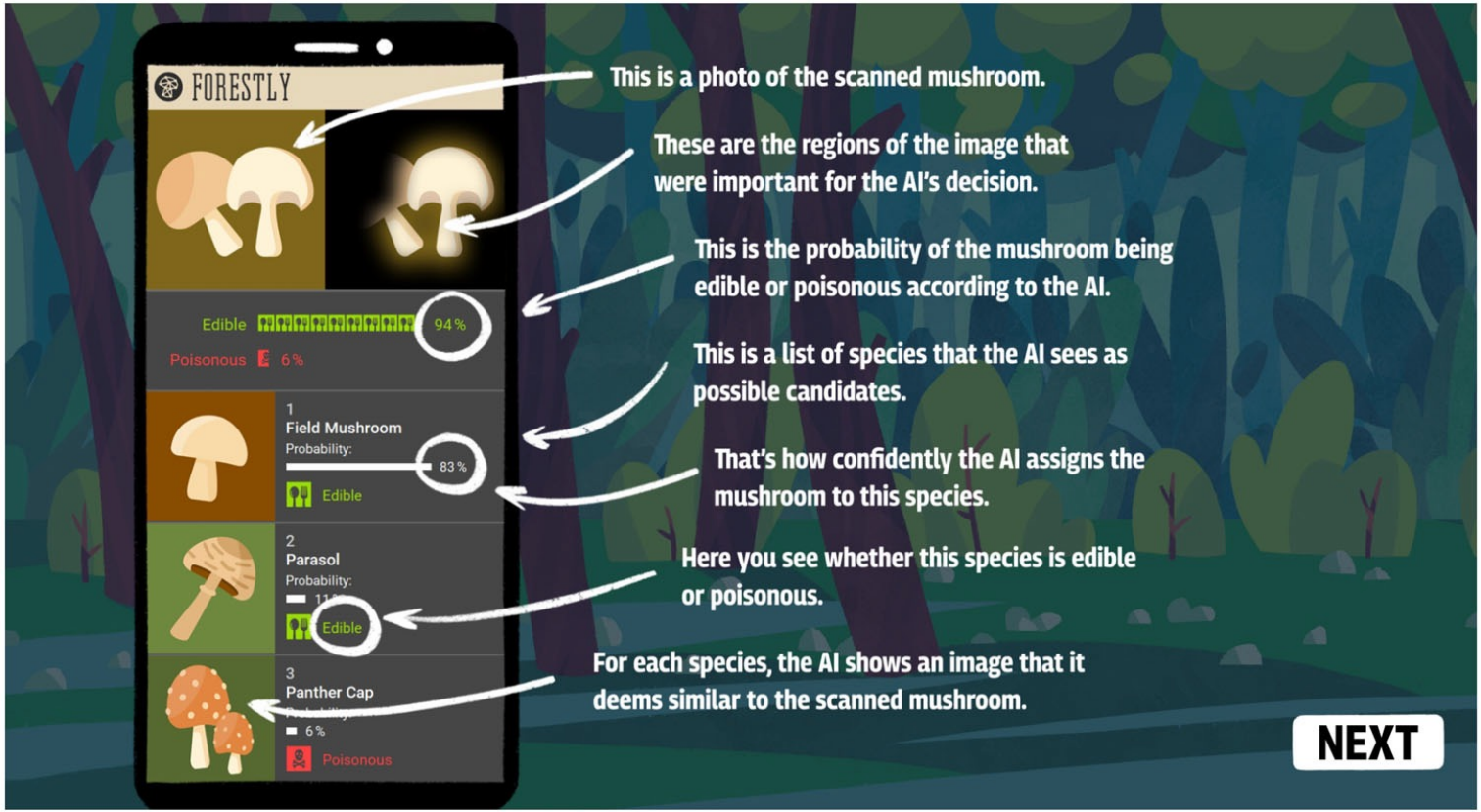


(3)

Review Sentiment Analysis [29]



Mushroom Classification [30]



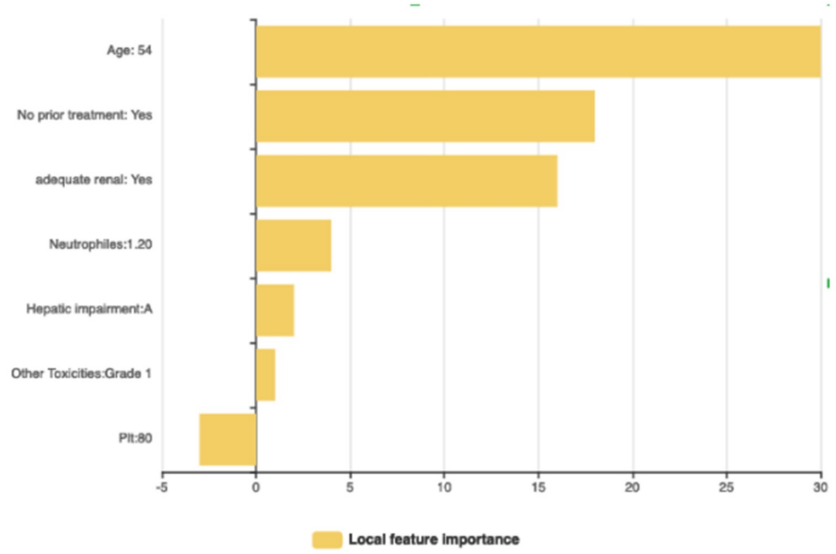
Prescription System [31]

Explanation Interface

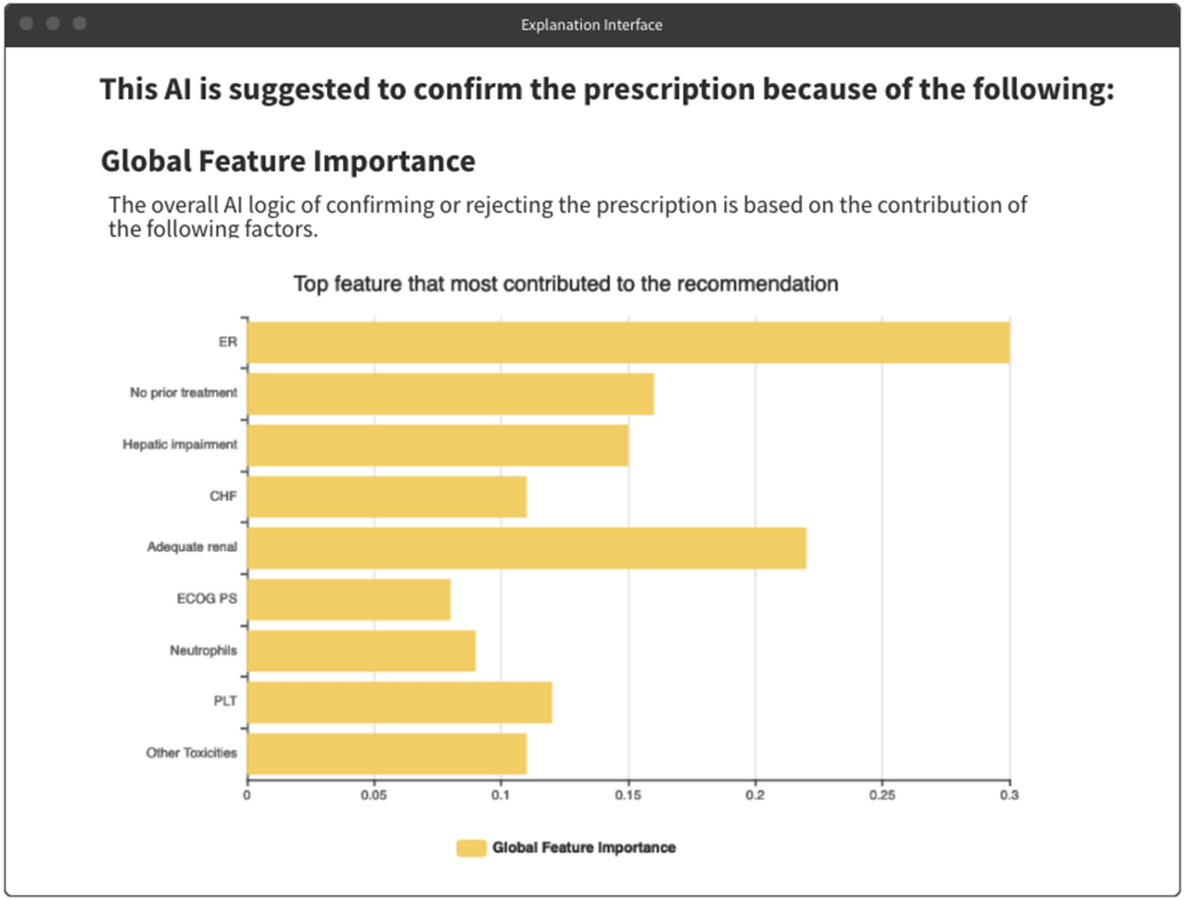
This AI is suggested to confirm the prescription because of the following:

Local Feature Importance

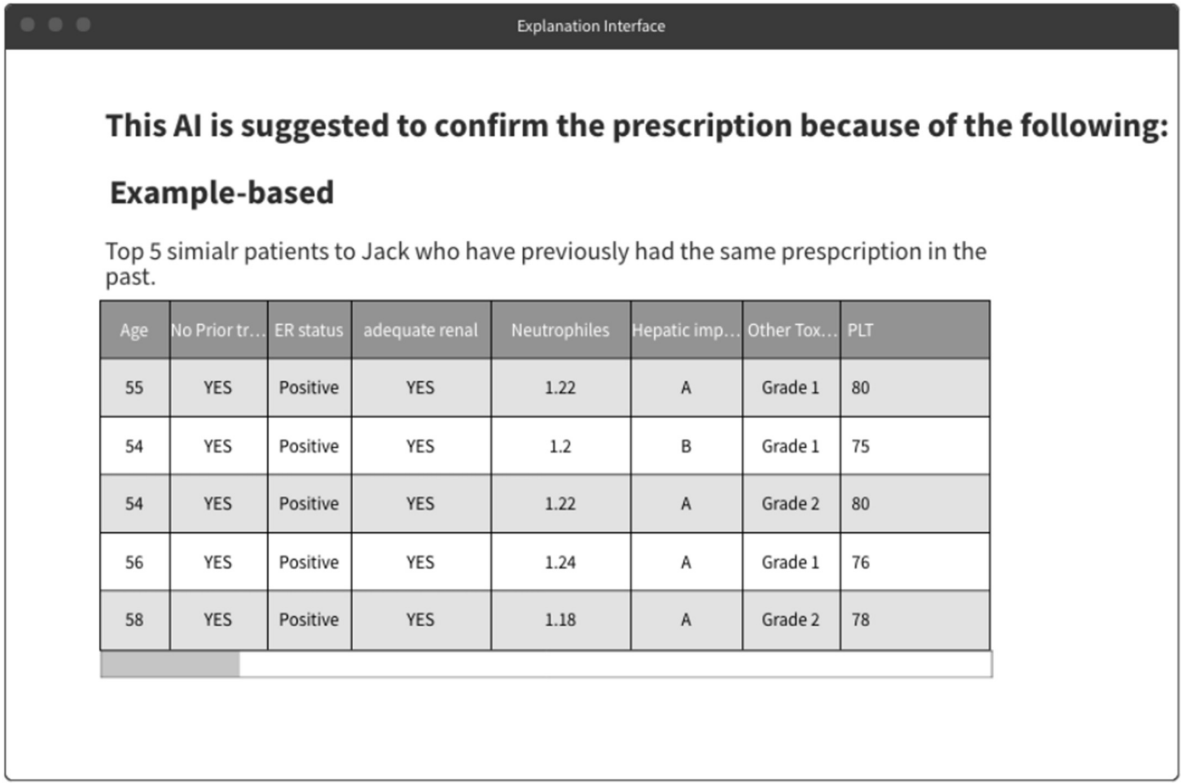
The AI think that the prescription should be confirmed because of the importance of each of the following factors.



Prescription System [31]



Prescription System [31]



Age	No Prior tr…	ER status	adequate renal	Neutrophiles	Hepatic imp…	Other Tox…	PLT
55	YES	Positive	YES	1.22	A	Grade 1	80
54	YES	Positive	YES	1.2	B	Grade 1	75
54	YES	Positive	YES	1.22	A	Grade 2	80
56	YES	Positive	YES	1.24	A	Grade 1	76
58	YES	Positive	YES	1.18	A	Grade 2	78

Prescription System [31]

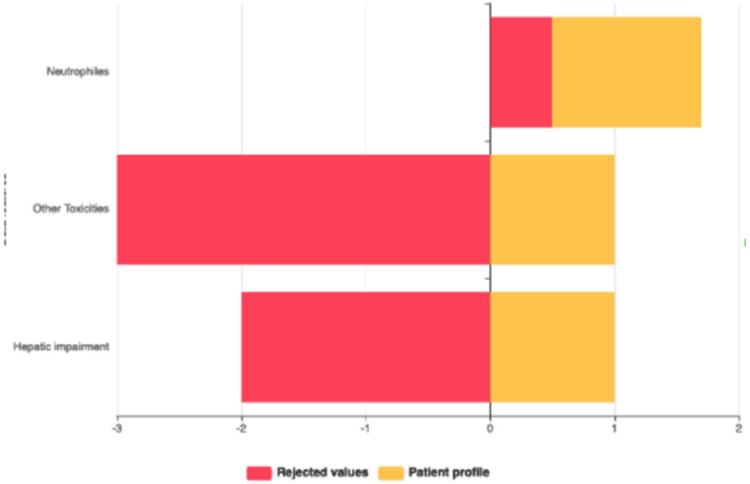
Explanation Interface

This AI is suggested to confirm the prescription because of the following:

Counterfactual Explanations

If the patient had one of following values, the prescription would have rejected:

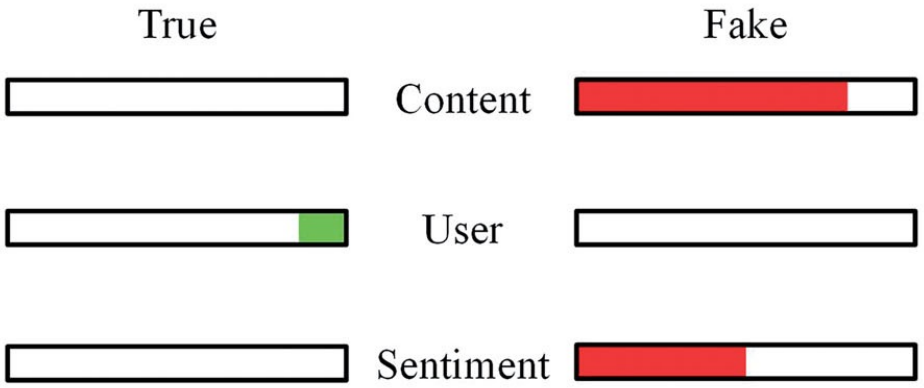
- Neutrophiles: 0.5
- Other toxicities: 4
- Plt:30



Fake News Detection [32]

ALERT

According to the common features of fake news captured by AI, this post is more likely to be fake news.
The current system accuracy is close to 90%, please browse carefully to avoid misjudgment.



A **large number** of writing styles common to fake news have been found in the news content and related comments.

No users who help spread misinformation have been detected.

Some negative sentiments have been captured from related comments, which is a common method used by fake news to incite users.

Pneumonia Diagnosis [33]

Patient 11

Does this patient have pneumonia?

Yes ☐ No ☐

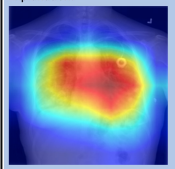
AI Suggestion

ALERT: confidence very high for a pneumonia prediction

Prediction: pneumonia

Confidence: 97%

Explanation:



Patient 11

Does this patient have pneumonia?

Yes ☐ No ☐

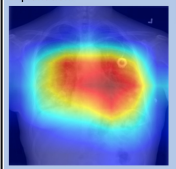
AI Suggestion

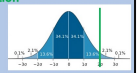
ALERT: confidence this high occurs only 3 out of 100 times for a pneumonia prediction

Prediction: pneumonia

Confidence: 97%

Explanation:





Patient 21

Does this patient have pneumonia?

Yes ☐ No ☐

AI Suggestion

CAUTION: confidence unusually low for a no-pneumonia prediction

Prediction: NO pneumonia

Confidence: 55%

Patient 21

Does this patient have pneumonia?

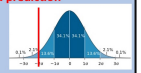
Yes ☐ No ☐

AI Suggestion

CAUTION: confidence this low occurs only 3 out of 100 times for a no-pneumonia prediction

Prediction: NO pneumonia

Confidence: 55%



Patient 20

Does this patient have pneumonia?

Yes ☐ No ☐

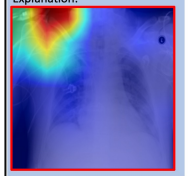
AI Suggestion

CAUTION: unusual explanation

Prediction: pneumonia

Confidence: 67%

Explanation:



Patient 20

Does this patient have pneumonia?

Yes ☐ No ☐

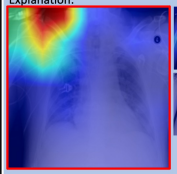
AI Suggestion

CAUTION: unusual explanation focused outside of lungs

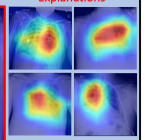
Prediction: pneumonia

Confidence: 67%

Explanation:



Typical Pneumonia Explanations



Patient 19

Does this patient have pneumonia?

Yes ☐ No ☐

AI Suggestion

CAUTION: model didn't see similar images in training

Prediction: NO pneumonia

Confidence: 79%

Patient 19

Does this patient have pneumonia?

Yes ☐ No ☐

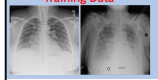
AI Suggestion

CAUTION: model didn't see images with small anatomies like this in training

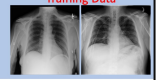
Prediction: NO pneumonia

Confidence: 79%

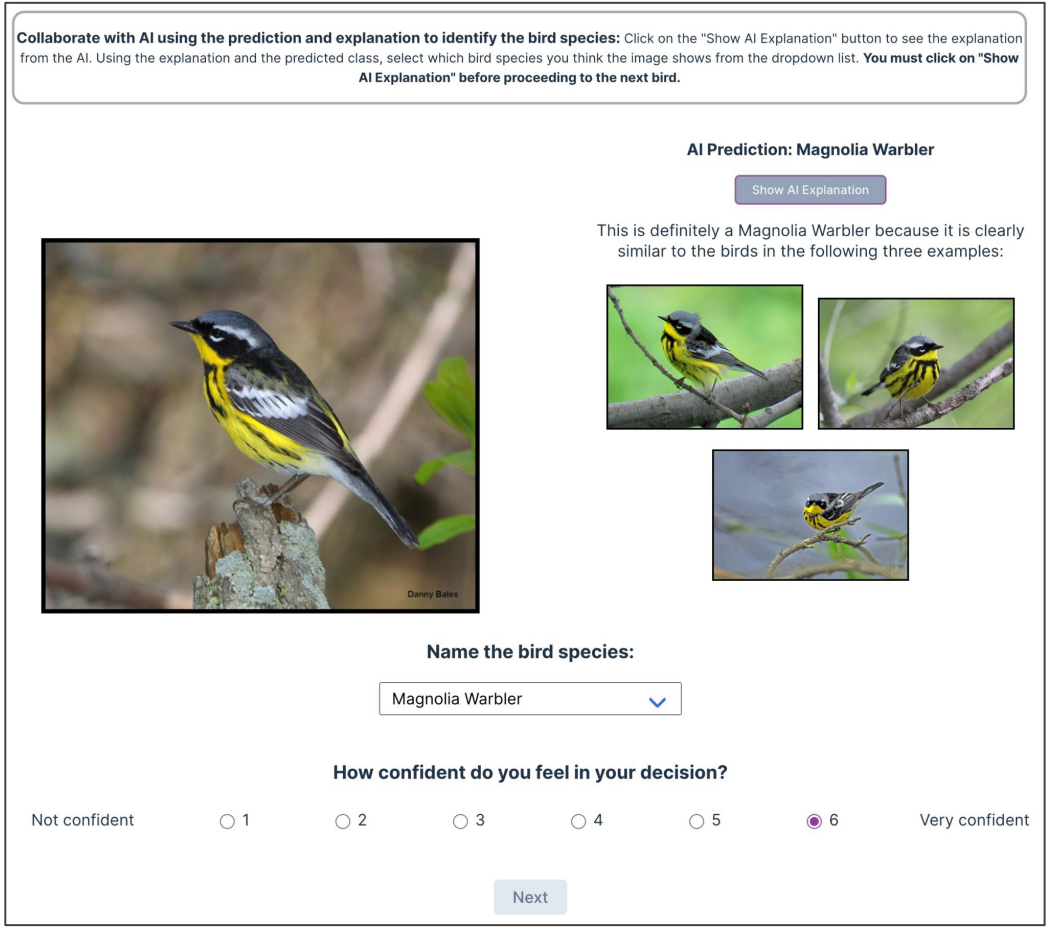
Typical Pneumonia Training Data



Typical No-Pneumonia Training Data



Bird Identification [34]



References

[1] Gunning, D., Vorm, E., Wang, Y., & Turek, M. (2021). **DARPA's explainable AI (XAI) program: A retrospective**. Authorea Preprints.

[2] Abdul, A., Vermeulen, J., Wang, D., Lim, B. Y., & Kankanhalli, M. (2018). **Trends and trajectories for explainable, accountable and intelligible systems: An hci research agenda**. In Proceedings of the 2018 CHI conference on human factors in computing systems (pp. 1-18).

[3] Miller, T. (2019). **Explanation in artificial intelligence: Insights from the social sciences**. Artificial intelligence, 267, 1-38.

[4] Doshi-Velez, F., & Kim, B. (2017). **A roadmap for a rigorous science of interpretability**. arXiv preprint arXiv:1702.08608, 2(1).

[5] Soremekun, E., Papadakis, M., Cordy, M., & Traon, Y. L. (2022). **Software fairness: An analysis and survey.** arXiv preprint arXiv:2205.08809.

[6] Diakopoulos, N. (2014). **Algorithmic accountability reporting: On the investigation of black boxes**.

[7] Cramer, H., Evers, V., Ramlal, S. et al. **The effects of transparency on trust in and acceptance of a content-based art recommender**. User Model User-Adap Inter 18, 455–496 (2008).

[8] Bellotti, V., & Edwards, K. (2001). **Intelligibility and accountability: human considerations in context-aware systems**. Human–Computer Interaction, 16(2-4), 193-212.

[9] Cheng, H. F., Wang, R., Zhang, Z., O'connell, F., Gray, T., Harper, F. M., & Zhu, H. (2019). **Explaining decision-making algorithms through UI: Strategies to help non-expert stakeholders**. In Proceedings of the 2019 CHI conference on human factors in computing systems (pp. 1-12).

[10] Cabrera, Á. A., Hohman, F., Lin, J., & Chau, D. H. (2018). **Interactive classification for deep learning interpretation**. arXiv preprint arXiv:1806.05660.

[11] Lee, M. K., & Baykal, S. (2017). **Algorithmic mediation in group decisions: Fairness perceptions of algorithmically mediated vs. discussion-based social division**. In Proceedings of the 2017 ACM CSCW (pp. 1035-1048).

[12] Martens, D., & Provost, F. (2014). **Explaining data-driven document classifications**. MIS quarterly, 38(1), 73-100.

[13] Stumpf, S., Rajaram, V., Li, L., Wong, W. K., Burnett, M., Dietterich, T., ... & Herlocker, J. (2009). **Interacting meaningfully with machine learning systems: Three experiments**. International journal of human-computer studies, 67(8), 639-662.

References

[14] Lim, B. Y. (2012). **Improving understanding and trust with intelligibility in context-aware applications** (Doctoral dissertation, Carnegie Mellon University).

[15] Becker, B., Kohavi, R., & Sommerfield, D. (2001). **Visualizing the simple Baysian classifier**. In Information visualization in data mining and knowledge discovery (pp. 237-249).

[16] Caragea, D., Cook, D., & Honavar, V. G. (2001). **Gaining insights into support vector machine pattern classifiers using projection-based tour methods**. In Proceedings of the seventh ACM SIGKDD international conference on Knowledge discovery and data mining (pp. 251-256).

[17] Jakulin, A., Možina, M., Demšar, J., Bratko, I., & Zupan, B. (2005). **Nomograms for visualizing support vector machines**. In Proceedings of the eleventh ACM SIGKDD international conference on Knowledge discovery in data mining (pp. 108-117).

[18] Tzeng, F. Y., & Ma, K. L. (2005). **Opening the black box-data driven visualization of neural networks** (pp. 383-390). IEEE.

[19] Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). **Grad-cam: Visual explanations from deep networks via gradient-based localization**. In Proceedings of the IEEE international conference on computer vision (pp. 618-626).

[20] Kim, B. (2015). **Interactive and interpretable machine learning models for human machine collaboration** (Doctoral dissertation, Massachusetts Institute of Technology).

[21] Krause, J., Perer, A., & Ng, K. (2016). **Interacting with predictions: Visual inspection of black-box machine learning models**. In Proceedings of the 2016 CHI conference on human factors in computing systems (pp. 5686-5697).

[22] Plate, T. A., Bert, J., Grace, J., & Band, P. (2000). **Visualizing the function computed by a feedforward neural network**. Neural computation, 12(6), 1337-1353.

[23] Olden, J. D., & Jackson, D. A. (2002). **Illuminating the "black box": a randomization approach for understanding variable contributions in artificial neural networks**. Ecological modelling, 154(1-2), 135-150.

[24] Liao, Q. V., & Varshney, K. R. (2021). **Human-centered explainable ai (xai): From algorithms to user experiences**. arXiv preprint arXiv:2110.10790.

References

[25] Clement, T., Kemmerzell, N., Abdelaal, M., & Amberg, M. (2023). **XAIR: A Systematic Metareview of Explainable AI (XAI) Aligned to the Software Development Process**. Machine Learning and Knowledge Extraction, 5(1), 78-108.

[26] Chen, V., Liao, Q. V., Wortman Vaughan, J., & Bansal, G. (2023). **Understanding the role of human intuition on reliance in human-AI decision-making with explanations**. Proceedings of the ACM on Human-computer Interaction, 7(CSCW2), 1-32.

[27] Natali, C., Famiglini, L., Campagner, A., La Maida, G. A., Gallazzi, E., & Cabitza, F. (2023). **Color shadows 2: Assessing the impact of xai on diagnostic decision-making**. In World Conference on Explainable Artificial Intelligence (pp. 618-629). Cham: Springer Nature Switzerland.

[28] Bertrand, A., Viard, T., Belloum, R., Eagan, J. R., & Maxwell, W. (2023). **On selective, mutable and dialogic xai: A review of what users say about different types of interactive explanations**. In Proceedings of the 2023 CHI conference on human factors in computing systems (pp. 1-21).

[29] Cau, F. M., Hauptmann, H., Spano, L. D., & Tintarev, N. (2023). **Effects of AI and logic-style explanations on users' decisions under different levels of uncertainty**. ACM Transactions on Interactive Intelligent Systems, 13(4), 1-42.

[30] Leichtmann, B., Hinterreiter, A., Humer, C., Streit, M., & Mara, M. (2024). **Explainable artificial intelligence improves human decision-making: results from a mushroom picking experiment at a public art festival**. International Journal of Human–Computer Interaction, 40(17), 4787-4804.

[31] Naiseh, M., Al-Thani, D., Jiang, N., & Ali, R. (2023). **How the different explanation classes impact trust calibration: The case of clinical decision support systems**. International Journal of Human-Computer Studies, 169, 102941.

[32] Chien, S. Y., Yang, C. J., & Yu, F. (2022). **XFlag: Explainable fake news detection model on social media**. International Journal of Human–Computer Interaction, 38(18-20), 1808-1827.

[33] Radensky, M., Burson, D., Bhaiya, R., & Weld, D. S. (2022). **Exploring How Anomalous Model Input and Output Alerts Affect Decision-Making in Healthcare**. arXiv preprint arXiv:2204.13194.

[34] Morrison, K., Spitzer, P., Turri, V., Feng, M., Kühl, N., & Perer, A. (2024). **The impact of imperfect XAI on human-AI decision-making**. Proceedings of the ACM on human-computer interaction, 8(CSCW1), 1-39.